

Educations

Ph.D. in Electrical and Electronic Engineering Supervised by Prof. Lihua Xie	Aug 2026 - present Nanyang Technological University (NTU), Singapore
Master Degree in Control Engineering and Automation Supervised by Prof. Tung Lam Nguyen	Jul 2024 - Jul 2026 Hanoi Univ of Science and Technology (HUST), Hanoi, Vietnam
Bachelor Degree in Control Engineering and Automation Supervised by Prof. Tung Lam Nguyen	Oct 2020 - Apr 2024 Hanoi Univ of Science and Technology (HUST), Hanoi, Vietnam

Work experiences

R&D Robotics Engineer at VinRobotics, Vingroup Full-time, supervised by Dr. Minh Nhat Vu	Sep 2025 - Jul 2026 Hanoi, Vietnam
Research Project Assistant at HUST Contract (Hybrid), supervised by PhD. Chinh Hoang Duc and Assoc.Prof.PhD. Tung Lam Nguyen	Oct 2021 - Dec 2026 Hanoi, Vietnam

Selected publications

- [7] **Vu, D. C.**, Nguyen, V. T., Nguyen, D. H., Nguyen, M. C., Tran, V. T., & Vu, M. N. (2026). Real-Time Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators. *IEEE RA-L*. [[pdf](#)]
- [6] Nguyen, T. T. H., **Vu, D. C.**, Pham, M. D., Nguyen, T. L., & Nguyen, T. V. A. (2026). Unifying hierarchical sliding mode control and control barrier function for tilt angle constraint of a ball-balancing robot. *IET Cyber-Systems and Robotics*. [[pdf](#)]
- [5] **Vu, D. C.**[†], Tran, S.[†], Nguyen, T. L., & Hoang, D. C. (2025). Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks. *Ocean Engineering*. [[pdf](#)] [†]equal contribution
- [4] **Vu, D. C.**, Nguyen, D. H., Vu, M. N., & Nguyen, T. L. (2025). Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations. *Ocean Engineering*, 340, 122142. [[pdf](#)]
- [3] Pham, M. D., **Vu, D. C.**, Nguyen, T. T. H., Nguyen, T. V. A., Vu, M. N., & Nguyen, T. L. (2025). CBFs-based model predictive control for obstacle avoidance with tilt angle limitation for ball-balancing robots. *IEEE Access*. [[pdf](#)]
- [2] **Vu, D. C.**, Nguyen, D. H., & Nguyen, T. L. (2024). A novel approach of consensus-based finite-time distributed sliding mode control for Stewart platform manipulators motion tracking. *Results in Engineering*. [[pdf](#)]
- [1] **Vu, D. C.**, Pham, M. D., Nguyen, T. T. H., Nguyen, T. V. A., & Nguyen, T. L. (2024). Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints. *International Journal of Robust and Nonlinear Control*. [[pdf](#)]

Honours & awards

NTU Research Scholarship	Nanyang Technological University (NTU), Singapore
Master, PhD Scholarship Programme	Vingroup Innovation Foundation (VINIF)
Best Thesis Defense Award	Hanoi University of Science and Technology

Academic activities

Invited review for *Nonlinear Dynamics, Ocean Engineering, IEEE Access, IEEE Robotics and Automation Letters (RA-L); International Conference on Intelligent Robots and Systems (IROS 2026), IEEE RO-MAN 2026, Humanoids 2026, International Conference on Robotics and Automation (ICRA 2027).*